Experiment No.: 6

Date of Performance:

Date of Submission:

Aim

Study of Bluetooth communication using mobile devices (pairing, range testing).

Required Equipment / Software

- Two Bluetooth-enabled smartphones, preferably Android devices.
- Built-in Bluetooth settings and a supported file-sharing/Bluetooth application.
- Measuring tape or meter scale.
- Suitable laboratory, corridor or open test area.
- A small test file such as a text file or image.

Theory

Bluetooth is a short-range wireless communication technology used to exchange data between nearby devices. It operates in the 2.4 GHz ISM (Industrial, Scientific and Medical) band and uses frequency-hopping techniques to improve robustness against interference.

Bluetooth device discovery allows a mobile device to identify nearby Bluetooth devices that are available for the required connection procedure. Pairing establishes a trusted relationship between two Bluetooth devices. Depending on the device and operating system, a passkey or confirmation code may be displayed during pairing.

After pairing, supported Bluetooth services can be used for activities such as file transfer, audio streaming and communication with peripherals. The available services depend on the mobile operating system and device.

Bluetooth range depends on transmit power, antenna characteristics, receiver sensitivity, obstacles, interference and the environment. As distance increases, the received signal generally becomes weaker and communication may become intermittent or fail.

Procedure

- Switched ON Bluetooth on both smartphones.
- Opened Bluetooth settings and enabled the required discovery/visibility option.
- On Smartphone A, searched for available Bluetooth devices.
- Identified Smartphone B from the list of discovered devices.
- Selected Smartphone B and initiated pairing.
- Confirmed the pairing code/passkey on both devices.
- Verified that the devices were successfully paired/connected.
- Sent a small test file from Smartphone A to Smartphone B using a supported Bluetooth/file-sharing method.
- Verified that the file was received successfully.

- For range testing, placed the two smartphones approximately 1 m apart and verified communication.
- Increased the distance in steps until communication became unreliable.
- At each distance, attempted a simple data-transfer or connection test and recorded the result.
- Determined the maximum distance at which reliable communication was achieved under the test conditions.
- Disconnected/unpaired the devices after completing the experiment.

Observation Table

Parameter	Observation
Device A	OPPO A74 5G
Device B	Realme 8i
Bluetooth status ON	Yes, enabled on both devices
Device discovery	Successful — Device B detected in scan list
Pairing status	Paired
Pairing code/passkey verified	Yes, confirmed on both devices
Test file name/type	sample.txt (text file)
Data transfer status	Successful

Bluetooth Range Testing

Sr. No.	Device Name	RSSI	Distance	Signal Strength
1	OPPO A74 5G	-57 dBm	~0.5 m	Excellent
2	Realme 8i	-56 dBm	~0.71 m	Excellent
3	Realme 12x 5g	-58 dBm	~0.89 m	Good
4	Generic	-69 dBm	~3.16 m	Good
5	Xiaomi	-80 dBm	~11.22 m	Fair / Weak

Approximate signal-quality guide: Very Strong (-30 to -50 dBm), Good (-51 to -67 dBm), Moderate (-68 to -75 dBm), Weak (-76 to -85 dBm), Very Weak (below -85 dBm). These ranges are approximate.

Result

Bluetooth communication between two mobile devices was successfully established through discovery and pairing. A test file was transferred, and the communication range was determined by increasing the distance between the devices and observing connection/data-transfer performance.

Conclusion

The experiment demonstrates basic Bluetooth communication using mobile devices. Pairing establishes a trusted relationship, while range testing shows that communication reliability generally decreases with increasing distance. Obstacles, interference, device characteristics and the surrounding environment affect practical Bluetooth range.

Viva Voce Questions

1. What is Bluetooth?

Bluetooth is a short-range wireless communication technology, based on the IEEE 802.15.1 standard, used to exchange data between nearby fixed or mobile devices over short distances using radio waves.

2. Which frequency band is used by Bluetooth?

Bluetooth operates in the 2.4 GHz ISM (Industrial, Scientific and Medical) band.

3. What is Bluetooth pairing?

Bluetooth pairing is the process of establishing a trusted relationship between two Bluetooth devices, typically involving discovery of the device and confirmation of a passkey or code, after which the devices can securely exchange data.

4. What factors affect Bluetooth range?

Bluetooth range is affected by transmit power, antenna characteristics, receiver sensitivity, obstacles (walls, furniture, the human body), interference from other 2.4 GHz devices, and the surrounding environment.

5. What is the 2.4 GHz ISM band?

The 2.4 GHz ISM (Industrial, Scientific and Medical) band is a range of radio frequencies set aside for unlicensed use by industrial, scientific and medical equipment, and is shared by technologies such as Bluetooth and Wi-Fi.

Marks & Signature

R1 (3 Marks)	R2 (5 Marks)	R3 (4 Marks)	R4 (3 Marks)	Total (15 Marks)	Signature